GOVERNMENT ENGINEERING COLLEGE, PALAKKAD

DEPARTMENT OF MECHANICAL ENGINEERING

PROJECT REPORT

ON

ANALYSIS OF ACKERMANN STEERING MECHANISM

SUBMITTED BY

BHARATH A	PKD24ME026
GAYATHRI J PISHARADY	PKD24ME031
MUHAMMAD SHAMEEM	PKD24ME052
SREEPAD T	PKD24ME065
SUDEEP C	PKD24ME067

Under the guidance of

Prof. Vidhun VR
Assistant Professor
Department of Mechanical Engineering

Academic Year: 2025–2026

Certificate

This is to certify that the project titled **"ANALYSIS OF ACKERMANN STEERING MECHANISM"** submitted by the students listed below to the Department of Mechanical Engineering, Government Engineering College, Palakkad, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Mechanical Engineering is a record of original work carried out under my supervision.

Student Team:

- BHARATH A PKD24ME026
- GAYATHRI J PISHARADY PKD24ME031
- MUHAMMAD SHAMEEM PKD24ME052
- SREEPAD T PKD24ME065
- SUDEEP C PKD24ME067

Guide: Mr. Vidhun VR

Place: Palakkad   Date: ____________________

DECLARATION

We hereby declare that the project report entitled **"ANALYSIS OF ACKERMANN STEERING MECHANISM"** is a record of the original work carried out by us in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Mechanical Engineering at Government Engineering College, Palakkad.

We further declare that this work has not been submitted, either in part or in full, for the award of any other degree or diploma.

Place: Palakkad Date: ____________________

Student Team:

BHARATH A	(sign)
GAYATHRI J PISHARADY	(sign)
MUHAMMAD SHAMEEM	(sign)
SREEPAD T	(sign)
SUDEEP C	(sign)

VISION AND MISSION

VISION AND MISSION OF THE COLLEGE

Vision

Excellence through the wings of science and technology.

Mission

To transform youth to talented engineers with creativity and integrity who can meet the technological challenges for the service of society.

VISION AND MISSION OF THE DEPARTMENT

Vision

To become a recognised center for imparting outstanding education, research, and consultancy in mechanical engineering.

Mission

- Impart quality education to mould successful engineers.
- Provide state-of-art infrastructure and research facilities.
- Strive for continuous improvement in overall quality of teaching, research and consultancy.

ABSTRACT

This project focuses on the design and development of a model of the Ackermann steering mechanism, a fundamental steering geometry used in automobiles to ensure efficient turning. The mechanism is based on the principle that all wheels of a vehicle must rotate about a common instantaneous center during a turn, thereby minimizing tire slip and improving handling performance.

In conventional steering systems, both front wheels often turn at equal angles, leading to inefficiencies such as tire wear and poor directional control. The Ackermann steering mechanism overcomes this limitation by allowing the inner wheel to turn at a greater angle than the outer wheel, ensuring smooth and accurate motion.

The project involves the design, fabrication, and testing of a small-scale mechanical model using simple materials such as wooden linkages and wheels. The expected outcome is a functional prototype that clearly demonstrates the working principle of the Ackermann steering system.

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INTRODUCTION

The steering system is a fundamental subsystem of any automobile, responsible for controlling the direction and stability of the vehicle during motion. Its primary function is to ensure that all wheels roll smoothly without slipping while executing a turn. In practical conditions, improper steering geometry leads to lateral skidding, increased tire wear, and loss of control. To address these issues, various steering mechanisms have been developed, among which the Ackermann steering mechanism is one of the most widely used due to its simplicity and effectiveness. This project focuses on analyzing the Ackermann steering mechanism through theoretical study, mathematical validation, and simulation to understand its real-world application.

Objectives

The main objective of this project is to design and analyze the Ackermann steering mechanism and understand its working principle. The project also aims to fabricate a simple prototype using basic materials to visually demonstrate steering geometry. Additionally, a Python-based simulation is developed to validate the Ackermann condition mathematically by computing inner and outer wheel angles. Through this combined approach, the project bridges theoretical concepts with practical implementation.

Need for Steering Geometry

When a vehicle turns, the inner and outer wheels follow different circular paths due to their different radii of rotation. The inner wheel travels along a smaller radius, while the outer wheel follows a larger radius. If both wheels are forced to turn at the same angle, it results in slipping, increased friction, and inefficient motion. This leads to faster tire wear, higher energy consumption, and reduced vehicle stability. Therefore, proper steering geometry is essential to ensure that each wheel rotates about a common instantaneous center, enabling smooth and efficient turning without lateral slipping.

- Tire skidding
- Increased friction
- Reduced efficiency

Types of Steering Mechanisms

Different steering mechanisms have been developed to achieve accurate wheel alignment during turning. The Ackermann steering mechanism is widely used in light vehicles due to its simplicity and ability to approximate ideal steering geometry. The Davis steering mechanism provides exact steering conditions using sliding pairs, but suffers from high friction and maintenance issues, making it impractical. The rack and pinion system is commonly used in modern vehicles for its compact design and quick response. The recirculating ball mechanism is typically used in heavy vehicles where higher load-handling capacity is required, though it is less responsive compared to rack and pinion systems.

1. **Ackermann Steering Mechanism**
   Ensures correct steering geometry and is widely used in light vehicles.
2. **Davis Steering Mechanism**
   Uses sliding pairs but has high friction and maintenance issues.
3. **Rack and Pinion**
   Most common in modern cars due to simplicity and precision.
4. **Recirculating Ball**
   Used in heavy vehicles.

Principle of Ackermann Steering Mechanism

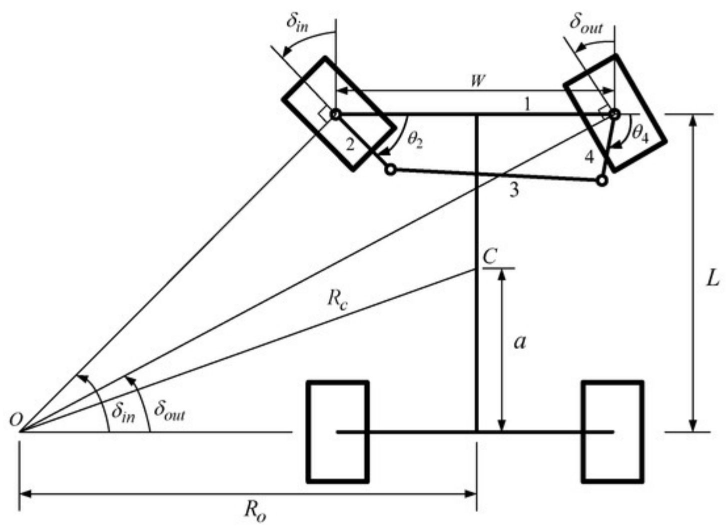

Figure 1: Ackermann steering mechanism

The Ackermann steering mechanism is based on the principle that all four wheels of a vehicle must rotate about a common instantaneous center during a turn. This ensures pure rolling motion without lateral slipping. When a vehicle turns, the rear wheels follow circular paths, and the front wheels must align such that their axes intersect at a single point on the rear axle line. To achieve this, the inner front wheel turns at a larger angle than the outer wheel. This difference in steering angles allows both wheels to move along concentric circular paths, ensuring smooth and efficient vehicle motion.

Construction of Ackermann Steering Mechanism

1. Front Wheels

- The two inclined wheels at the top are the front wheels.
- They rotate at different angles during turning:
  - The inner wheel turns more.
  - The outer wheel turns less.
- This difference is the core principle of Ackermann steering.

2. Steering Arms

- The short arms attached to each front wheel are the steering arms.
- They are fixed to the wheel hubs (steering knuckles).
- Their main functions are:
  - Transfer motion from linkage to wheels.
  - Help create the correct steering angles.

3. Tie Rod

- The connecting rod between the two steering arms is the tie rod.
- Its functions are:
  - Connect both steering arms.
  - Ensure coordinated movement of both wheels.
- When one arm moves, the tie rod forces the other arm to move accordingly.

4. Steering Linkage

- The combination of:
  - Two steering arms
  - One tie rod

  forms a trapezoidal linkage.
- This trapezoidal geometry is the heart of Ackermann steering.
- It ensures that both wheels turn about a common instantaneous centre.

5. Rear Wheels

- The two wheels at the bottom are the rear wheels.
- They:
  - Do not steer.
  - Follow the front wheels.

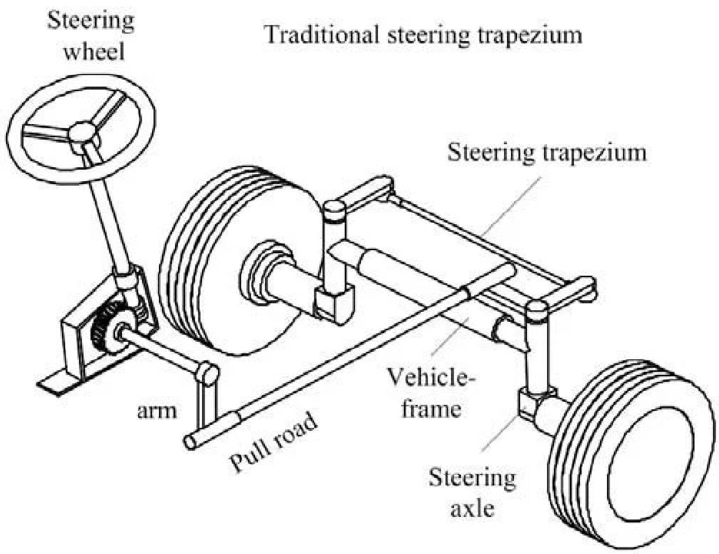

Figure 2: Parts

Mathematical Condition

The Ackermann steering condition establishes the relationship between the inner and outer wheel angles, track width, and wheelbase. It is expressed as:

$$\cot\theta - \cot\phi = \frac{b}{l}$$

where is the inner wheel angle, is the outer wheel angle, b is the track width, and l is the wheelbase. This equation ensures that the axes of all wheels intersect at a common instantaneous center. Satisfying this condition minimizes slipping and guarantees correct steering geometry.

Working

When the driver turns the steering wheel, the motion is transmitted through the steering system to the tie rod. This causes the steering arms to rotate, which in turn changes the orientation of the front wheels. Due to the trapezoidal arrangement of the linkage, the inner wheel rotates through a larger angle compared to the outer wheel. This allows both wheels to align with their respective circular paths during a turn. The axes of the wheels intersect at a common instantaneous center, ensuring smooth rolling motion and reducing tire wear.

Derivation

The derivation of the Ackermann steering condition is based on the geometry of circular motion. For a vehicle turning with radius R, the inner and outer wheel angles are given by:

$$\tan\theta = \frac{l}{R - \frac{b}{2}}$$

$$\tan\phi = \frac{l}{R + \frac{b}{2}}$$

Taking reciprocals and subtracting the equations leads to:

$$\cot\theta - \cot\phi = \frac{b}{l}$$

This derivation confirms the geometric condition required for proper steering alignment and ensures that all wheels rotate about a common center.

SIMULATION DEVELOPMENT

The simulation is developed to validate the Ackermann steering condition using computational methods. A curved road path is defined mathematically, and the vehicle's motion along this path is simulated. The curvature at each point is calculated, which is then used to determine the turning radius. Based on this radius, the inner and outer wheel angles are computed using Ackermann equations. The simulation visually represents the vehicle movement and dynamically updates the steering angles, providing a clear understanding of how the mechanism behaves under different conditions.

- Define road path
- Calculate curvature
- Compute angles

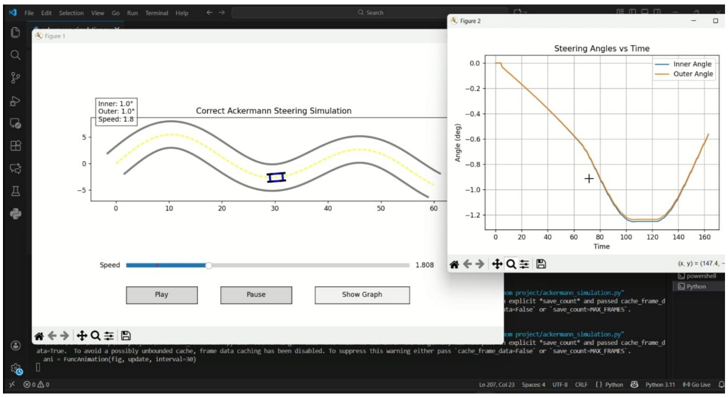

Figure 1.1: Simulation development

Python Code

The following Python code implements a simulation of the Ackermann steering mechanism to validate the theoretical relationship between inner and outer wheel angles. The program generates a curved road path and calculates the vehicle's orientation and curvature at each point. Based on this curvature, the steering angles for the inner and outer wheels are computed using Ackermann geometry. The simulation visually represents the vehicle motion and dynamically updates the wheel orientations. Additionally, the code records the variation of inner and outer steering angles over time and plots them in a graph, enabling clear analysis of their relationship and confirming that the inner wheel consistently turns at a greater angle than the outer wheel during motion.

The simulation was implemented using Python with NumPy and Matplotlib libraries. The program computes curvature from a predefined path and applies Ackermann equations to determine inner and outer steering angles. The angles are updated dynamically and plotted against time for analysis. The complete code is provided in Appendix A.

Algorithm

1) Calculate path coordinates
2) Compute orientation
3) Calculate curvature
4) Apply Ackermann equations
5) Animate motion

```
theta_inner = atan(wheelbase/(R - track_width/2))
theta_outer = atan(wheelbase/(R + track_width/2))
```

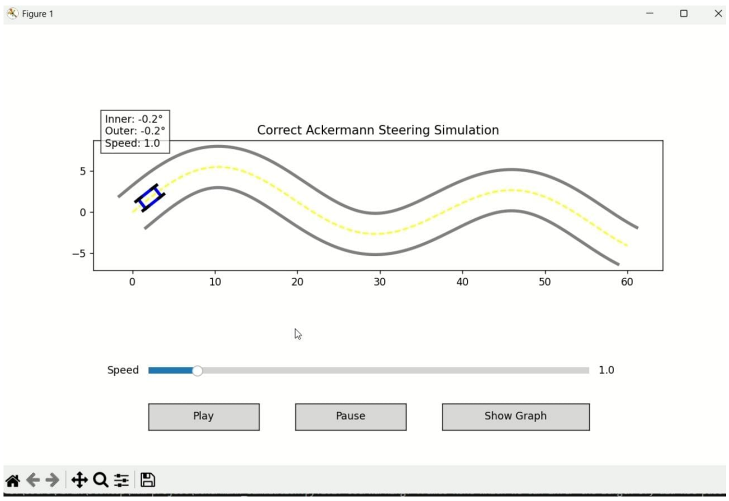

Figure 1.2: ackermann simulation

RESULTS AND ANALYSIS

The results obtained from the simulation confirm the theoretical behavior of the Ackermann steering mechanism. The vehicle follows the defined curved path smoothly, and the steering angles vary continuously based on the curvature of the path. The inner wheel consistently exhibits a larger steering angle compared to the outer wheel. The graphical output clearly shows the variation of steering angles over time, validating the mathematical relationship between them. The simulation effectively demonstrates the correctness of the Ackermann condition.

Observations

From the simulation and analysis, it is observed that the inner wheel always turns at a greater angle than the outer wheel, which is essential for proper steering. The motion of the vehicle remains smooth without any indication of lateral slipping. The coordination between the wheels confirms that the linkage design satisfies the Ackermann condition.

- Inner wheel angle is greater
- Smooth turning without slipping

Graph Analysis

The graph plotted between steering angle and time illustrates the variation of inner and outer wheel angles during motion. The inner wheel angle curve consistently lies above the outer wheel angle curve, indicating a larger turning angle. The smooth variation of both curves reflects stable steering behavior. This graphical representation provides clear validation of the theoretical relationship and helps in understanding the dynamic behavior of the steering mechanism.

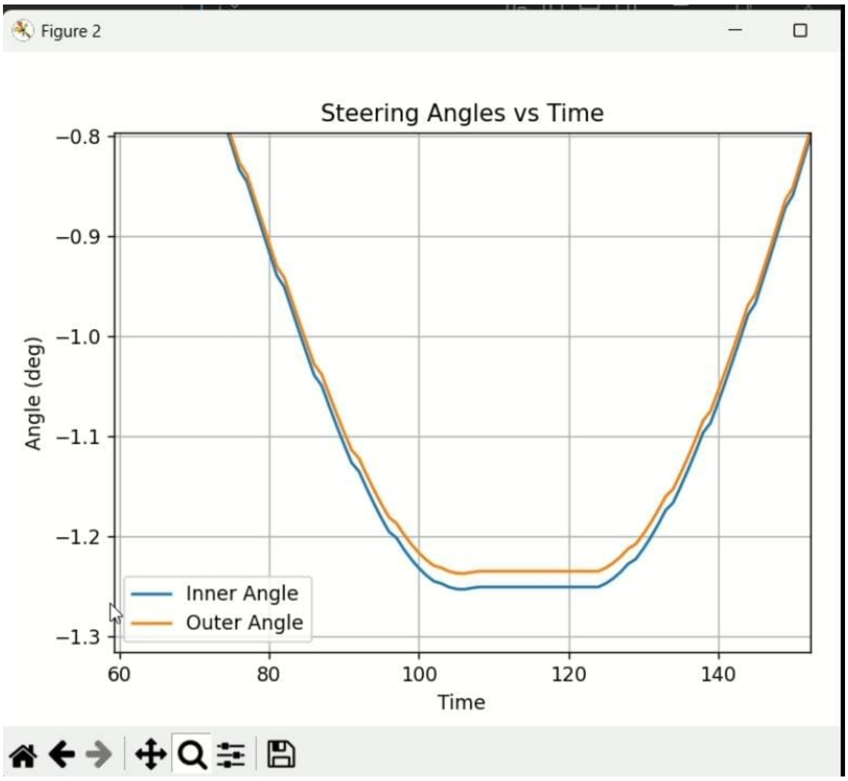

Figure 2.1: Steering Angles vs Time

The plotted graph clearly demonstrates the variation of inner and outer steering angles as the vehicle follows a curved path. It is evident that the inner wheel consistently exhibits a greater steering angle than the outer wheel, which aligns with the theoretical requirements of the Ackermann steering mechanism. This difference in angles ensures that both wheels trace concentric circular paths about a common instantaneous center, thereby minimizing lateral slipping. The smooth and continuous nature of the curves indicates stable steering behavior and validates the effectiveness of the implemented simulation in accurately representing real-world steering dynamics.

ADVANTAGES AND LIMITATIONS

Advantages and Limitations

Advantages

- **Correct Steering Geometry:** It ensures that all four wheels of a vehicle roll about a common instantaneous centre during turning. This minimizes lateral slipping of the tires.
- **Reduced Tire Wear:** Since the inner and outer wheels turn at different angles, unnecessary skidding is avoided, which significantly reduces tire wear and increases tire life.
- **Smooth Turning:** The mechanism provides smooth and natural turning motion, improving driving comfort and vehicle handling.
- **Simple Construction:** Compared to more complex steering systems, Ackermann steering is relatively simple in design and easy to manufacture.
- **High Efficiency at Low Speeds:** It performs very well at low speeds and in moderate turning conditions, making it ideal for passenger cars and light vehicles.

Limitations

- **Not Perfect at High Speeds:** At high speeds, dynamic effects like tire deformation and centrifugal forces cause deviation from ideal Ackermann geometry.
- **Limited Accuracy in Real Conditions:** Road conditions, suspension movement, and tire flexibility reduce the accuracy of the ideal steering angles.
- **Not Suitable for All Vehicles:** Heavy vehicles, racing cars, or vehicles requiring precise high-speed handling may use modified or alternative steering geometries.
- **Complexity in Precise Design:** Although simple in concept, achieving exact Ackermann geometry requires precise alignment and design calculations.
- **Wear and Maintenance Issues:** Linkages and joints can wear out over time, affecting steering accuracy and requiring maintenance.

PROTOTYPE

The prototype developed in this project is a simplified physical model designed to demonstrate the working principle of the Ackermann steering mechanism. It primarily focuses on illustrating the correct arrangement of steering arms and tie rods to achieve the required geometry. Although the model does not operate dynamically like a real vehicle, it effectively shows how different steering angles are produced. This makes it a useful educational tool for understanding steering linkage behavior and validating theoretical concepts.

Figure 4.1: Prototype developed

Materials

The prototype is constructed using simple and easily available materials such as wood for the base structure, nuts and bolts for joints, and toy wheels for representation. These materials are chosen to keep the model cost-effective while still effectively demonstrating the mechanism.

CONCLUSION

The project successfully demonstrates the working and analysis of the Ackermann steering mechanism. It establishes the relationship between wheelbase, track width, and steering angles, and verifies the theoretical condition through simulation. The results confirm that the inner and outer wheels turn at different angles to achieve smooth and efficient motion. Despite minor practical limitations, the Ackermann mechanism remains a reliable and widely used steering solution. Overall, the project enhances understanding of mechanical design, kinematics, and real-world automotive systems.

APPENDIX

Appendix A: Python Simulation Code

```
# Full Python code exactly as used in the simulation
#(import matplotlib
matplotlib.use('TkAgg')
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation
from matplotlib.widgets import Slider, Button
import math
#----------------------------
# PARAMETERS
#----------------------------
wheelbase = 2.4
track_width = 1.4
speed = 1.0
running = True
frame_idx = 0.0
# data for graph
inner_hist = []
outer_hist = []
#----------------------------
# ROAD
#----------------------------
t = np.linspace(0, 60, 800)
road_x = t
road_y = 4*np.sin(t/6) + 2*np.sin(t/12)
dx = np.gradient(road_x)
dy = np.gradient(road_y)
angles = np.arctan2(dy, dx)
#----------------------------
# FIGURE
```

```
#----------------------------
fig, ax = plt.subplots(figsize=(10,6))
plt.subplots_adjust(bottom=0.3)
ax.set_title("Correct␣Ackermann␣Steering␣Simulation")
# road
lane = 2.5
norm = np.sqrt(dx**2 + dy**2)
nx =-dy / norm
ny = dx / norm
ax.plot(road_x + lane*nx, road_y + lane*ny, 'gray',
linewidth=3)
ax.plot(road_x- lane*nx, road_y- lane*ny, 'gray',
linewidth=3)
ax.plot(road_x, road_y, 'yellow', linestyle='--')
# car + wheels
car_body, = ax.plot([], [], 'blue', linewidth=3)
wheels = [ax.plot([], [], 'black', linewidth=3)[0] for _ in
range
(4)]
info = ax.text(0.02, 0.95, "", transform=ax.transAxes,
bbox=dict(facecolor='white', alpha=0.7))
#----------------------------
# TRANSFORM
#----------------------------
def transform(x, y, a, lx, ly):
X = x + lx*np.cos(a)- ly*np.sin(a)
Y = y + lx*np.sin(a) + ly*np.cos(a)
return X, Y
#----------------------------
# CAR + WHEEL
#----------------------------
def get_car(x,y,a):
L,W = wheelbase, track_width
pts = [(-L/2,-W/2),(L/2,-W/2),(L/2,W/2),
(-L/2,W/2),(-L/2,-W/2)]
return np.array([transform(x,y,a,px,py) for px,py in pts])
def get_wheel(x,y,a,s):
pts = [(-0.4,0),(0.4,0)]
return np.array([transform(x,y,a+s,px,py) for px,py in pts])
#----------------------------
```

```
# ANIMATION
#----------------------------
def update(frame):
global frame_idx
if running:
frame_idx += speed
if frame_idx >= len(road_x)-2:
frame_idx = 0
i = int(frame_idx)
x,y = road_x[i], road_y[i]
angle = angles[i]
# curvature
if i > 5:
k = (angles[i]- angles[i-5]) / 5
else:
k = 0
if abs(k) < 1e-4:
R = 1e6
else:
R = 1/k
#----------------------------
# TRUE ACKERMANN
#----------------------------
if abs(R) > 1e5:
left = right = 0
else:
theta_inner = math.atan(wheelbase/(R- track_width/2))
theta_outer = math.atan(wheelbase/(R + track_width/2))
if k > 0: # left turn
left = theta_inner
right = theta_outer
else:
# right turn
left = theta_outer
right = theta_inner
# store data
inner_hist.append(math.degrees(left))
outer_hist.append(math.degrees(right))
# draw car
body = get_car(x,y,angle)
```

```
car_body.set_data(body[:,0], body[:,1])
# wheel local positions (correct layout)
L,W = wheelbase, track_width
wheel_local = [
(-L/2,-W/2), # rear left
(-L/2, W/2), # rear right
( L/2,-W/2), # front left
( L/2, W/2) # front right
]
for j,(lx,ly) in enumerate(wheel_local):
wx, wy = transform(x,y,angle,lx,ly)
if j == 2:
steer = left
elif j == 3:
steer = right
else:
steer = 0
w = get_wheel(wx,wy,angle,steer)
wheels[j].set_data(w[:,0], w[:,1])
info.set_text(
f"Inner:␣{math.degrees(left):.1f} \n"
f"Outer:␣{math.degrees(right):.1f} \n"
f"Speed:␣{speed:.1f}"
)
return [car_body] + wheels + [info]
ani = FuncAnimation(fig, update, interval=30)
#---------------------------
# UI
#---------------------------
ax_speed = plt.axes([0.2,0.2,0.6,0.03])
speed_slider = Slider(ax_speed, "Speed", 0.5, 5,
valinit=speed)
def update_speed(val):
global speed
speed = speed_slider.val
speed_slider.on_changed(update_speed)
# buttons
ax_play = plt.axes([0.2,0.08,0.15,0.06])
ax_pause = plt.axes([0.4,0.08,0.15,0.06])
ax_graph = plt.axes([0.6,0.08,0.2,0.06])
```

```
btn_play = Button(ax_play,"Play")
btn_pause = Button(ax_pause,"Pause")
btn_graph = Button(ax_graph,"Show␣Graph")
def play(event):
global running
running = True
def pause(event):
global running
running = False
def show_graph(event):
plt.figure()
plt.plot(inner_hist,label="Inner␣Angle")
plt.plot(outer_hist,label="Outer␣Angle")
plt.legend()
plt.title("Steering␣Angles␣vs␣Time")
plt.xlabel("Time")
plt.ylabel("Angle␣(deg)")
plt.grid()
plt.show()
btn_play.on_clicked(play)
btn_pause.on_clicked(pause)
btn_graph.on_clicked(show_graph)
ax.set_aspect('equal')
plt.show())
```

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